

Datasheet: Agriculture-Vision

1. Motivation

For what purpose was the dataset created? Was there a specific task in mind? Was there a specific gap that needed to be filled? Please provide a description.

The dataset was created to encourage and advance research in agricultural pattern analysis and semantic segmentation. Furthermore, it has been created to facilitate the development of efficient algorithms for precision agriculture, which is critical for mitigating losses and optimizing seasonal yields.

The dataset was created to encourage and advance research in agricultural pattern analysis and semantic segmentation. Furthermore, it has been created to facilitate the development of efficient algorithms for precision agriculture, which is critical for mitigating losses and optimizing seasonal yields. The Agriculture-Vision dataset was developed to bridge the gap between general computer vision and the unique complexities of precision agriculture by providing an unprecedented resolution of 10 cm/px, multispectral imagery (RGB and NIR) and precise annotations from professional agronomists with a strict quality assurance process.

Who created the dataset (e.g., which team, research group) and on behalf of which entity (e.g., company, institution, organization)?

Mang Tik Chiu¹, Xingqian Xu¹, Yunchao Wei¹, Zilong Huang¹, Alexander Schwing¹, Robert Brunner¹, Hrant Khachatryan², Hovnatan Karapetyan², Ivan Dozier², Greg Rose², David Wilson², Adrian Tudor², Naira Hovakimyan^{2,1}, Thomas S. Huang¹, Honghui Shi^{3,1}.

These researchers are members of the following cooperating entities:

- ¹UIUC (University of Illinois Urbana-Champaign)
- ²Intelinair
- ³University of Oregon

Who funded the creation of the dataset? If there is an associated grant, please provide the name of the grantor and the grant name and number.

Unknown to the authors of the datasheet.

Any other comments?

None.

2. Composition

What do the instances that comprise the dataset represent (e.g., documents, photos, people, countries)? Are there multiple types of instances (e.g., movies, users, and ratings; people and interactions between them; nodes and edges)? Please provide a description.

Aerial farmland images, collected throughout the growing seasons at numerous farming locations in the US, where several important field patterns were annotated by agronomy experts.

How many instances are there in total (of each type, if appropriate)?

The dataset was initially proposed in 2020, consisting of 21,061 images and it was expanded in 2021, reaching a total of 94,986 images.

Does the dataset contain all possible instances or is it a sample (not necessarily random) of instances from a larger set? If the dataset is a sample, then what is the larger set? Is the sample representative of the larger set (e.g., geographic coverage)? If so, please describe how this representativeness was

validated/verified. If it is not representative of the larger set, please describe why not (e.g., to cover a more diverse range of instances, because instances were withheld or unavailable).

The Agriculture-Vision dataset was constructed using a specific sampling and preprocessing methodology to make it suitable for modern deep learning architectures. The original, unprocessed farmland images are extremely large (the largest field image has 33571×24351 pixels). To manage this scale, the images were partitioned into cropped annotations with a window size of 512×512 pixels. But isn't representative because this windowing process was realized with a specific focus on the expert labels so only patches containing relevant agricultural patterns were retained, while windows consisting purely of background were discarded.

What data does each instance consist of? "Raw" data (e.g., unprocessed text or images) or features? In either case, please provide a description.

Each instance consists of images with a resolution of 10 cm/px, size of 512×512 pixels and multi-spectral data comprising aligned RGB and NIR channels. Each image also has a boundary map and a mask. The boundary map indicates the region of the farmland, and the mask indicates valid pixels in the image. Regions outside of either the boundary map or the mask are not evaluated. Furthermore, this dataset contains nine types of annotations about types of field anomalies.

Is there a label or target associated with each instance? If so, please provide a description.

Yes, each instance is associated with binary masks of nine types of field anomalies: *drydown*, *nutrient deficiency*, *weed cluster*, *water*, *endrow*, *double plant*, *waterway*, *storm damage*, and *planter skip*. These annotations were labeled by five annotators trained by expert agronomists through commercial software.

Is any information missing from individual instances? If so, please provide a description, explaining why this information is missing (e.g., because it was unavailable). This does not include intentionally removed information, but might include, e.g., redacted text.

Everything is included. No data is missing.

Are relationships between individual instances made explicit (e.g., users' movie ratings, social network links)? If so, please describe how these relationships are made explicit.

Yes, the relationships are primarily spatial and geographic because of the way the data was sampled from the original large-scale field images and they originate from the same farmland.

Are there recommended data splits (e.g., training, development/validation, testing)? If so, please provide a description of these splits, explaining the rationale behind them.

Yes, there's a split in 56,944/18,334/19,708 train/val/test images. The researchers first randomly split the 3,432 original large-scale farmland images using a 60/20/20% ratio so each 512×512 cropped image was then assigned to the same split as its parent farmland image. This approach ensures no cropped patches from the same physical plantation appear in more than one split.

Are there any errors, sources of noise, or redundancies in the dataset? If so, please provide a description.

There is an inherent source of noise regarding the manual annotations. Because these patterns were labeled manually, achieving absolute precision at the pixel level is challenging. Furthermore, some NIR images are found seems to have overflow problems, as shown in the Image 1.

Is the dataset self-contained, or does it link to or otherwise rely on external resources (e.g., websites, tweets, other datasets)?

The dataset is self-contained.

Does the dataset contain data that might be considered confidential (e.g., data that is protected by legal privilege or by doctor–patient confidentiality, data that includes the content of individuals' non-public communications)? If so, please provide a description.

No.

Does the dataset contain data that, if viewed directly, might be offensive, insulting, threatening, or might otherwise cause anxiety? If so, please describe why.

No.

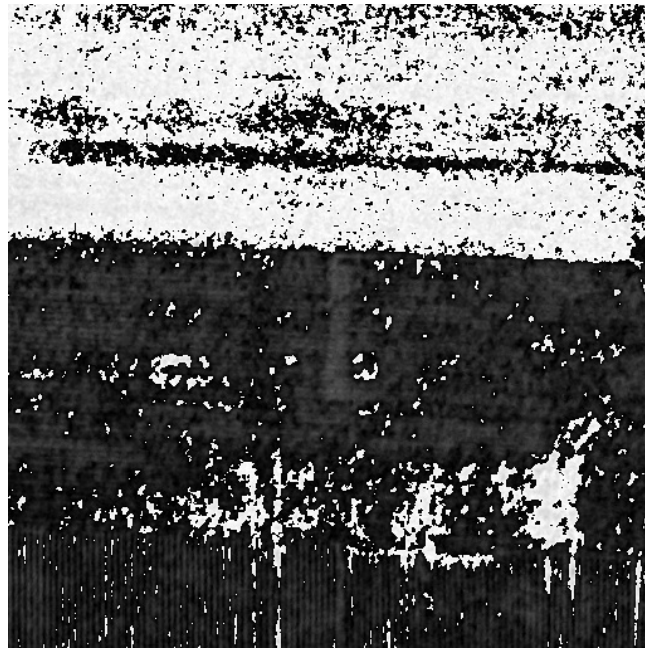


Figure 1: NIR image with overflow problem.

3. Collection Process

How was the data associated with each instance acquired? Was the data directly observable (e.g., raw text, movie ratings), reported by subjects (e.g., survey responses), or indirectly inferred/derived from other data (e.g., part-of-speech tags, model-based guesses for age or language)? If the data was reported by subjects or indirectly inferred/derived from other data, was the data validated/verified? If so, please describe how.

The image data was directly observed and captured through aerial photography of thousands of farmland fields across the United States and the labels for each instance were obtained through manual expert analysis rather than being automatically generated or reported by subjects.

What mechanisms or procedures were used to collect the data (e.g., hardware apparatuses or sensors, manual human curation, software programs, software APIs)? How were these mechanisms or procedures validated?

The capture setup evolved across the collection years as follows:

- **2017 Farmland Images:** Captured using two aligned *Canon SLR* cameras, where one captured standard RGB images and the other captured only the NIR channel.
- **2018 Farmland Images:** The NIR, Red, and Green (NRG) channels were captured using two *Nikon D850* cameras to enable the 10 cm/px resolution. Custom filters were deployed to capture near-infrared instead of the blue channel. In parallel, separate Blue channel images were captured using one *Nikon D800E* at 20 cm/px resolution, which were subsequently scaled up to align with the corresponding NRG images.
- **2019 Farmland Images:** Captured using a proprietary *Wide Area Multi-Spectral System (WAMS)* commonly used for remote sensing. The WAMS captures all four channels simultaneously at 10 cm/px resolution.

The validation of these physical mechanisms is unknown to the authors of this datasheet.

If the dataset is a sample from a larger set, what was the sampling strategy (e.g., deterministic, probabilistic with specific sampling probabilities)?

The sampling strategy is deterministic because it follows a fixed logic to select which data is included in the final dataset. Rather than capturing a random sample of the entire landscape, the methodology applies a strict retention rule with only 512x512 pixel windows containing anomalies are kept, while any windows consisting solely of background are systematically discarded.

Who was involved in the data collection process (e.g., students, crowdworkers, contractors) and how were they compensated (e.g., how much were crowdworkers paid)?

Unknown to the authors of this datasheet.

Over what timeframe was the data collected? Does this timeframe match the creation timeframe of the data associated with the instances (e.g., recent crawl of old news articles)? If not, please describe the timeframe in which the data associated with the instances was created.

The data was collected over three years, between 2017 and 2019. Subsequently the dataset was compiled, annotated and made available in January 2020.

Were any ethical review processes conducted (e.g., by an institutional review board)? If so, please provide a description of these review processes, including the outcomes, as well as a link or other access point to any supporting documentation.

No, there is no ethical review process.

4. Preprocessing/cleaning/labeling

Was any preprocessing/cleaning/labeling of the data done (e.g., discretization or bucketing, tokenization, part-of-speech tagging, SIFT feature extraction, removal of instances, processing of missing values)? If so, please provide a description. If not, you may skip the remaining questions in this section.

While the 2017 images were already stored in standard 8-bit values, the 2018 and 2019 farmland images were initially captured in camera-raw format. To ensure consistency, these images underwent a normalization process where each of the four spectral channels was clipped based on the 5th and 95th percentiles. So the channels were separated and stored into a regular RGB image and an additional single-channel NIR image, and stored as two JPG images.

Was the “raw” data saved in addition to the preprocessed/cleaned/labeled data (e.g., to support unanticipated future uses)? If so, please provide a link or other access point to the “raw” data.

No, the dataset consists only of the preprocessed and cropped 512x512 patches extracted during the sampling process.

Is the software that was used to preprocess/clean/label the data available? If so, please provide a link or other access point.

No.

Any other comments?

No.

5. Uses

Has the dataset been used for any tasks already? If so, please provide a description.

Yes, the dataset serves as the foundational benchmark for the annual Agriculture-Vision Prize Challenge hosted at the CVPR Workshops. It is specifically used for the development and benchmarking of semantic segmentation models designed to detect specific agricultural anomalies.

Is there a repository that links to any or all papers or systems that use the dataset? If so, please provide a link or other access point.

No, a centralized repository does not exist, however, papers and systems using the dataset can be found in academic databases such as IEEE Xplore (Example 1, Example 2, Example 3).

What (other) tasks could the dataset be used for?

In addition to semantic segmentation, the dataset can also be used for image classification of the nine anomalies or binary classification (Healthy x Anomalous).

Is there anything about the composition of the dataset or the way it was collected and preprocessed/cleaned/labeled that might impact future uses? For example, is there anything that a dataset consumer might need to know to avoid uses that could result in unfair treatment of individuals or groups (e.g., stereotyping, quality of service issues) or other risks or harms (e.g., legal risks, financial harms)? If so, please provide a description. Is there anything a dataset consumer could do to mitigate these risks or harms?

Yes, this is a dataset created mainly with soybeans and corn fields from the United States (Illinois and Iowa). Models trained on this data might have problems with different geographic regions or different types of crops resulting in financial harms for farmers.

Are there tasks for which the dataset should not be used? If so, please provide a description.

The dataset should not be used as an application for farmlands outside the United States because the data is geographically biased toward the country and industrial farming techniques.

Any other comments?

No.

6. Distribution

Will the dataset be distributed to third parties outside of the entity (e.g., company, institution, organization) on behalf of which the dataset was created? If so, please provide a description.

Yes, the dataset is used for competitions every year since 2020 and is broadly distributed. Was intentionally created to be a resource publicly available for the global research community.

How will the dataset will be distributed (e.g., tarball on website, API, GitHub)? Does the dataset have a digital object identifier (DOI)?

No, it is distributed through a dedicated GitHub repository ([link here](#)) with download options using aws cli or huggingface. Although there isn't a specific DOI for the raw data, the dataset is commonly recognized and cited using the original publication's DOI: <https://doi.org/10.48550/arXiv.2001.01306><https://doi.org/10.48550/arXiv.2001.01306>

When will the dataset be distributed?

This dataset is available online from 2020, and every year is distributed in CVPR Challenges and Workshops.

Will the dataset be distributed under a copyright or other intellectual property (IP) license, and/or under applicable terms of use (ToU)? If so, please describe this license and/or ToU, and provide a link or other access point to, or otherwise reproduce, any relevant licensing terms or ToU, as well as any fees associated with these restrictions.

Yes, the dataset is distributed under a specific "Terms and Conditions for Use of Dataset" agreement by IntelinAir, Inc. It is a non-exclusive, royalty-free, limited license for internal, non-commercial academic purposes only. The full terms can be accessed at <https://www.agriculture-vision.com/dataset-terms>. There are no fees associated with these restrictions.

Have any third parties imposed IP-based or other restrictions on the data associated with the instances? If so, please describe these restrictions, and provide a link or other access point to, or otherwise reproduce, any relevant licensing terms, as well as any fees associated with these restrictions.

There are no explicit IP-based restrictions imposed by third parties.

Do any export controls or other regulatory restrictions apply to the dataset or to individual instances? If so, please describe these restrictions, and provide a link or other access point to, or otherwise reproduce, any supporting documentation.

No.

Any other comments?

None.

7. Maintenance

Who will be supporting/hosting/maintaining the dataset?

Intelinair is the lead entity responsible for the dataset's creation and ongoing management. On github it is distributed and maintained by SHI Labs.

How can the owner/curator/manager of the dataset be contacted (e.g., email address)?

support@intelinair.com (we can't find a contact related to the dataset for 2020's page but found a reference ([link](#)

here).

Is there an erratum? If so, please provide a link or other access point.

No.

Will the dataset be updated (e.g., to correct labeling errors, add new instances, delete instances)? If so, please describe how often, by whom, and how updates will be communicated to dataset consumers (e.g., mailing list, GitHub)?

Yes, the dataset can be enhanced for new challenges. If there is any update to the dataset, this will be communicated on the challenge's page for that year.

If the dataset relates to people, are there applicable limits on the retention of the data associated with the instances (e.g., were the individuals in question told that their data would be retained for a fixed period of time and then deleted)? If so, please describe these limits and explain how they will be enforced.

Does not relate.

Will older versions of the dataset continue to be supported/hosted/maintained? If so, please describe how. If not, please describe how its obsolescence will be communicated to dataset consumers.

This is the older version and there is no document talking about this but this is a dataset used for challenges that may be improved with more data sometime, as it was in 2024.

If others want to extend/augment/build on/contribute to the dataset, is there a mechanism for them to do so? If so, please provide a description. Will these contributions be validated/verified? If so, please describe how. If not, why not? Is there a process for communicating/distributing these contributions to dataset consumers? If so, please provide a description.

There is no specific mechanism but we can find an extended dataset with new data for the challenge of 2024. (link here)

Any other comments?

None.